Introduction to Operations Research, Deterministic Models

Lecture 1: Introduction of Mathematical Programming

Department of Industrial Engineering, Tsinghua University

Instructor: Junyi Liu

Sep 10, 2024

Course Information

- Time: Tuesday, 9:50 - 10:35, 10:40 - 11:35, 11:30 - 12:15
- Email: junyiliu@tsinghua.edu.cn (will do my best to respond to the emails within 12 hours)
- Office Hours: every Tuesday afternoon, 15:00 - 16:00 or by email-appointment
- Office: Shunde Building, Room 220A
- Teaching Assistant: Ziyi Jia, jzy23@mails.tsinghua.edu.cn

Announcements will be posted through **Tsinghua Web Learning** and **Wechat group chat**, please check them regularly

Course information

OR Textbooks:

- *Wayne L. Winston*, **Operations Research, Applications and Algorithms**, 4th Edition
- *Hillier, and Lieberman*, **Introduction to Operations Research**, 11th Edition.

Content:

- Linear Programming: modeling, algorithm, and theory
- Integer Programming: modeling, algorithm, and implementation
- Nonlinear Programming: optimality conditions and algorithm
- Dynamic Programming: modeling and algorithm

Grading: Homework(15 %), Midterm exam(30 %, week 8), Final exam(45 %), Participatation (10 %)

Homework Requirements

- Homeworks will be announced through **Tsinghua Web Learning** after the class with the due time.
- Homework assignments must be submitted before the class in person or through **Tsinghua Web Learning**.
- Late homework submissions are not accepted under any circumstances. The lowest score will not be considered in your final grade. Start your homework assignments early.
- For the exercises of numerical experiments, it is required to submit both codes and the report including the process, results, and analysis in detail.
- You are encouraged to discuss homework assignments with your classmates. However, everyone should finish the homework on his/her own. There is no tolerance of plagiarization. If you have gained help from classmates, TA or other resources, please cite the sources in your homework.

In today's class, we will introduce

- Introduction of OR
  - origin of OR
  - nature of OR
  - model formulation
- Introduction of Linear Programming (LP)
  - definitions
  - standard model
  - solutions
  - assumptions
  - examples

The origin of OR

The need to allocate the available resources to the various activities effectively for the organization provides the environment for the emergence of Operations Research (OR).

In World War II, there was an urgent need to allocate scarce resources to the various military operations and to the activities within each operation in an effective manner.

The interests in OR are shifted from military to industry. The study of optimization received very little attention until 1947, when Dantzig developed modern linear optimization; www.informs.org/Explore/History-of-O.R.-Excellence/Oral-Histories/George-Dantzig

The simplex method by Dantzig and the computer revolution play a key role for the rapid growth of OR.

OR now is embracing the new discipline of analytics, data science, machine learning, etc.

The nature of OR

- Observe the problem and gather all relevant data
- Formulate a mathematical model that abstracts the essence of the real problem with hypothesis
- Test the hypothesis via model validations and make modifications if needed such that the model is a precise representation of the problem
- Solve the problem and provide decisions to the decision-maker

Model formulation

Components

- decision variables $\mathbf{x} \in \mathbb{R}^n$
- objective function $f(\mathbf{x})$: profit, diversity, market share, stability, etc.
- constraints: resource, stability, physical
- parameters: constants in the objective function, and constraints.

In textbook problems, the parameters are usually given ahead.

But in reality, determining parameter values requires gathering relevant data and appropriate **data analytics**.

Because of the uncertainty about the true value of the parameter, it is important to analyze how the solution derived from the model would change if the value assigned to the parameter were changed to other plausible values (**sensitivity analysis**)

Model formulation

A general optimization problem:

Minimize	$f(\mathbf{x})$		(objective function)
subject to	$g_i(\mathbf{x}) \leq 0,$	for $i = 1, \ldots, m$	(inequality constraints)
	$h_i(\mathbf{x}) = 0,$	for $i = 1, \ldots, \ell$	(equality constraints)
	$\mathbf{x} \in X$		(set constraint)

Trade-off between precision and tractability:
model approximations are made to ensure model tractability (solvable) and also model validity (model predicts the relative effects of the alternative courses of action with sufficient accuracy)

Model solutions

optimal solutions: ideal, but not practical in many cases

stationary solutions: satisfying optimality conditions in nonlinear programming

suboptimal solutions: usually is achieved by heuristic procedures, when the time or cost for finding an optimal solution is too large.

Why study the solution methods?

- For applications, it is common to apply software solvers to solve the problem
- For research, it is important to solve the (new) problem more **efficiently, correctly, robustly**, by accounting the structure and properties of new problems
- Postoptimalty analysis is also important: optimality to the model, not the real problem.
- What-if analysis: what would happen to the optimal solution if different assumptions are made about future conditions

Mathematical Programming V.S. Heuristic Algorithms

Extendable:

mathematical programming deals with general problems with particular structures;

therefore extendable when extra constraints/costs are added

Interpretable:

why the solution is good?

how good the solution is?

what is the price of constraints?

what if the parameters/data are changed?

An LP example: the Wyndor problem

WYNDOR GLASS CO. produces high-quality glass products, including windows and glass dorrs. It has three plants. Aluminum frames and hardware are made in Plant 1, wood frames are made in Plant 2, and Plant 3 produces the glass and assembles the products.

Because of declining earnings, top management has decided to revamp the company's product line. Unprofitable products are being discontinued, releasing production capacity to launch two new products having large sales potential:

Product 1: An 8-foot glass door with aluminum framing
Product 2: A 4 $\times 6$ foot double-hung wood-framed window

The marketing division has concluded that the company could sell as much of either product as could be produced. Product 1 requires some of the production capacity in Plants 1 and 3, but none in Plant 2. Product 2 needs only Plants 2 and 3. However, because both products would be competing for the same production capacity in Plant 3, it is not clear **which mix of the two products would be most profitable.**

An LP example: the Wyndor problem

Define the problem: determine what the production rates (continuous variables) should be for the two products in order to maximize the total profit, subject to the restrictions imposed by the limited production capacities in three plants.

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Gather data:

- number of hours of production time available per week in each plant for new products;
- number of hours of production time used in each plant for Product 1 and 2 per batch.
- profit per batch produced of each new product, recognizing there is no substantial costs incurred to initiate the production and marketing for new products.

	Production Time per Batch, Hours		
	Product		
Plant	1	2	Production Time Available per Week, Hours
1	1	0	4
2	0	2	12
3	3	2	18
Profit per batch	$3,000	$5,000	

An LP example: the Wyndor problem

Model formulation:

- decision variables: x_1 = number of batches of product 1 produced per week; x_2 = number of batches of product 2 produced per week
- objective: total profit per week in thousands of dollars $Z = 3x_1 + 5x_2$
- constraints: $x_1 \leq 4$, $2x_2 \leq 12$, $3x_1 + 2x_2 \leq 18$

$$
\begin{aligned}
\text{maximize} \quad & Z = 3x_1 + 5x_2 \\
\text{subject to} \quad & x_1 \leq 4 \\
& 2x_2 \leq 12 \\
& 3x_1 + 2x_2 \leq 18 \\
& x_1 \geq 0, x_2 \geq 0
\end{aligned}
$$

a classic **resource-allocation problem**

Graphical method

A two-dimensional LP can be solved by the graphical method.

Wyndor problem

$$
\begin{aligned}
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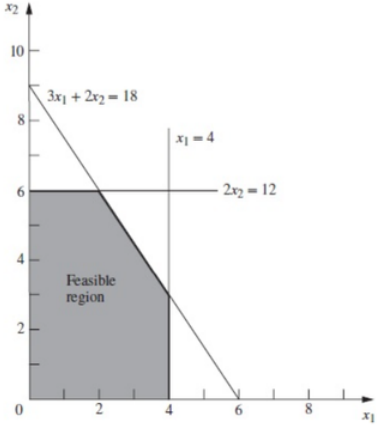

The LP model

Definition: Linear Programming is a technique for the optimization of a linear objective function, subject to linear equality and linear inequality constraints. Its feasible region is a set defined as the intersection of finitely many half spaces, each of which is defined by a linear inequality.

General form of LP:

$$
\begin{aligned}
\text{maximize} \quad & Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n \\
\text{subject to} \quad & a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq b_1 \\
& a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \leq b_2 \\
& \quad \vdots \\
& a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \leq b_m \\
& x_1 \geq 0, x_2 \geq 0, \ldots x_n \geq 0
\end{aligned}
$$

- LP problems of mid-size or large size can be efficiently solved by well-established commercial solvers.
- It is necessary to study the essence of LP solution methods, and theory behind, which provides the basis for post-optimality analysis

Is it an LP?

$$
\begin{aligned}
\text{maximize} \quad & Z = 3\,(x_1)^2 + 5x_2 \\
\text{subject to} \quad & x_1 \leq 4 \\
& 2x_2 \leq 12 \\
& 3x_1 + 2x_2 \leq 18 \\
& x_1 \geq 0, x_2 \geq 0
\end{aligned}
$$

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& 2x_2 \leq 12 \\
& 3\,x_1 + 2x_2 < 18 \\
& x_1 \geq 0, x_2 \geq 0
\end{aligned}
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& x_1 \geq 0, x_2 \geq 0
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$$

Is it an LP?

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\begin{aligned}
\text{maximize} \quad & Z = 3\,x_1 + 5x_2 \\
\text{subject to} \quad & x_1 \leq 4 \\
& 2x_2 \leq 12 \\
& x_1/x_2 \leq 4 \\
& x_1 \geq 0, x_2 \geq 1
\end{aligned}
$$

Solutions to the LP model

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- There could be no feasible solutions (infeasible LPs), or unbounded objectives, which lead to no optimal solutions
- There could also be multiple optimal solutions

Assumptions of LP

Proportionality

The contribution of each activity to the value of the objective function Z is proportional to the level of the activity x_j, as represented by the $c_j x_j$ term in the objective function.

Similarly, the contribution of each activity to the left-hand side of each functional constraint is proportional to the level of the activity x_j, as represented by the $a_{ij} x_j$ term in the constraint.

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The proportional assumption could be violated when:

- there is start-up costs for the production facilities
- there is an increasing marginal return because of economies of scale through the use of more efficient machinery, quantity discounts for large purchases of raw materials, the learning-curve effect of workers, etc.
- there is a decreasing marginal return due to the growing advertising costs for extra sales

These would lead to mixed-integer programs and nonlinear programs.

Assumptions of LP

Additivity

Every function in a linear programming model (whether the objective function or the constraint functions) is the sum of the individual contributions of the respective activities.

Th additivity assumption is violated when the two products are complementary (competitive) such that the joint profit is more (less) than the sum of their individual profits.

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Divisibility

Decision variables are allowed to have any values, including fractional, noninteger values.

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Certainty

The value assigned to each parameter of a LP model is assumed to be a known constant.

The certainty assumption is seldomly satisfied in practice, which necessitates sensitivity analysis.

Diet problem

Cindy's diet requires the food from a food group of four types of food: Brownie, chocolate ice cream, Cola and Pineapple cheesecake.

Each brownie costs 5 dollars, each scoop of chocolate ice cream costs 2 dollars, each bottle of cola costs 1.5 dollars, and each piece of pineapple cheesecake costs 8 dollars.

Cindy wants to know under the requirement of ingesting at least 500 calories, 6 oz of chocolate, 10 oz of sugar, and 8 oz of fat each day from such a food group, how much of each type of food should be eaten daily at the lowest cost?

Nutritinoal Values for Diet

Types of food	Calories	Chocolate	Sugar	Fat
Brownie	400	3	2	2
Chocolate ice cream	200	2	2	4
Cola	150	0	4	1
Pineapple	500	0	4	0

Diet problem

A version of diet problem with a more realistic list of foods and nutritional requirements waas one of the first LPs to be solved by computer.

Stigler (1945) proposed a problem in which 77 types of food were available and 10 nutritional requirements (vitamin A, vitamin C and so on) had to be satisfied. However, such a plan does not meet the minimum standard of tastiness.

Currently, one could further add constraints on tastiness and other various requirements (low fat, popularity, time cost) by incorporating the menu-ratings and recommendations in social media.

Work-scheduling problem

A post officer requires different numbers of full-time employees on different days of the week.

The number of employees required on each day is given in the following table

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
17	13	15	19	14	16	11

Each full time worker must work five consecutive days and receive two days off.

Determine the work-scheduling with the smallest number of full-time workers in total.

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Define x_i=number of workers beginning work on day i for $i = 1, 2, 3, 4, 5, 6, 7$.

Application: production process problem

Rylon manufactures Brute and Chanelle perfumes. Raw material can be purchased at $3/lb$. Processing 1 lb of raw material requires 1 lab hour. Each lb of raw material yields 3 oz of regular Brute and 4 oz of regular Chanelle.

Regular Brute can be sold for $7/oz$. Regular Chanelle can be sold for $6/oz$. Rylon has 6000 lab hours and purchase up to 4000 lb of raw material.

Rylon has the option of further processing regular Brute and regular Chanelle to produce Luxury Brute (sold at $18/oz$) and Luxury Chanelle (sold at $14/oz$). Each oz of regular Brute processed further requires an additional 3 hours of lab hour and $4 extra processing cost and yields 1 oz of Luxury Brute. Each oz of regular Chanelle processed further requires an additional 2 hours of lab hour and $4 extra processing cost and yields 1 oz of Luxury Chanelle.

What is the best production plan to maximize profits?

Application: production process problem

Decision variables:

$x_1 =$ number of oz of regular Brute sold annually
$x_2 =$ number of oz of Luxury Brute sold annually
$x_3 =$ number of oz of regular Chanelle sold annually
$x_4 =$ number of oz of Luxury Chanelle sold annually
$x_5 =$ number of oz of raw materials purchased annually